

Tangential A_4 Background Laws at Fixed Calibration

Codex

ABSTRACT: We study background graphs relating simultaneous A_4 zero-form and calibrated $\mathbb{Z}_2^{(1)}$ one-form symmetries in a bosonic 1 + 1-dimensional lattice gauge model on unoriented surfaces. Both symmetries act microscopically, and the graph records symmetry fractionalization. Our class consists of source-covariant finite-depth conjugates of one gapped Hamiltonian with regulated bulk, boundary, source, and defect tensors. Characteristic descent for each rank-two bundle reduces the pointed law 2-type to the discrete components 0 and the binary tetrahedral class ω . The identity $H^1(BA_4; \mathbb{Z}_2) = 0$ removes the mixed tangent term; $H^2(BA_4; \mathbb{Z}_2) = \mathbb{Z}_2 \langle \omega \rangle$ supplies the ordinary component. On $\mathbb{RP}^2$, the equality $w_1^2 = w_2 = \omega(A) \neq 0$ holds, and a finite-coupling partition function separates the two flux sectors. Tangent-dependent background graphs are obstructed within this characteristic-law class. A UV-to-IR transmutation requires a model whose symmetry content changes with scale.

KEYWORDS: Generalized Symmetries, Discrete Symmetries, Topological Defects, Lattice Gauge Theory, Group Cohomology

Contents

1	Introduction	2
2	The microscopic class	3
2.1	Tangential structure, gauge field, and matter	3
2.2	Exact one-form symmetry and the controlled Hamiltonian	4
2.3	Definition of the certified class	5
2.4	Symmetry inventory	5
3	Regulator and gluing	6
3.1	State sum and gauge transformations	6
3.2	Hamiltonian transfer matrix	7
3.3	Refinement and boundary gluing	7
3.4	Generating source and defect tensors	9
4	Laws from the defect OPE	9
5	Classification	10
5.1	Stackification and the relative mapping space	11
5.2	A_4 cohomology and the relative groups	11
5.3	Fixed-calibration classification theorem	12
6	The cocycle and its defects	13
6.1	Binary tetrahedral cocycle	13
6.2	Wall junctions and orientation defects	14
7	Partition functions and the $\mathbb{RP}^2$ test	14
7.1	Exact partition comparison	14
7.2	Finite-coupling flux sectors	15
7.3	Simultaneously nonzero tangent and ordinary classes	15
7.4	A nonzero ordinary fractionalization completion	16
8	Scope	16
9	Conclusion	17
A	Finite certificates	17

1 Introduction

Seiberg and Seifnashri define symmetry transmutation as a dynamical passage from a UV symmetry G to an IR symmetry K [1]. A background relation between symmetries acting at one scale records symmetry fractionalization. Ordinary cohomological examples assign a higher-form background to the zero-form field A :

$$B_{p+1} = \beta(A), \quad [\beta] \in H^{p+1}(BG, K). \quad (1.1)$$

For $p = 1$, β is the class of a central extension of G by K . This language also describes generalized symmetries and their anomalies [2, 3]. Appendix B of ref. [1] suggests tangent-bundle classes as an ingredient in higher-form, beyond-cohomology transmutation for $p \geq 2$. A related degree-two kinematic question arises for two simultaneous symmetries.

A tangent-dependent background law has the form

$$B_{p+1} = \Phi(A, w_1(TM), w_2(TM), \dots), \quad (1.2)$$

A pointed universal class is tangential when it retains tangent data after source-gauge homotopy in a fixed background coordinate. Pointing removes tangent terms present at trivial G source. A universal cochain homotopy can identify a tangent representative with an ordinary class. Invertible phases classified through cobordism pose a separate problem [4].

Our microscopic model lives on an arbitrary $O(2)$ surface. Each link carries a $\mathbb{Z}_2$ gauge qubit. A neutral four-state spin on each site carries the faithful permutation action of A_4 . The hard Gauss law leaves an electric one-form symmetry, from which we select the calibrated two-object subgroup

$$K = \mathbb{Z}_2^{(1)}, \quad \mathcal{U}_K(x) = X_\ell^g, \quad (1.3)$$

where the link ℓ crosses the spatial point x .

The theorem concerns Hamiltonians equipped with their regulated source and defect data. Its allowed Hamiltonians are

$$H_T = V H_* V^{-1}. \quad (1.4)$$

The circuit V has bounded depth and range. It fixes the calibrated symmetries and intertwines every bulk, boundary, source, and defect tensor for formal (A, E, B) , where E is an arbitrary rank-two $O(2)$ bundle. These objects are source-covariant presentations of one gapped model.

To extract the part induced by G , we take the Picard difference between the graph $\Phi(A, E)$ and its value at trivial A_4 source:

$$\Psi(A, E) = \Phi(A, E) - \Phi(1, E). \quad (1.5)$$

Both evaluations use the same formal bundle E and independent B -coordinate. The defect calibration stays fixed. Equation (1.5) removes pure w_1^2 and w_2 origins and retains any mixed class.

For graphs that descend to homotopy-invariant characteristic laws, we find

$$\tau_{\leq 2} \text{Map}((BA_4 \times BO(2), \{*\} \times BO(2)), (B^2\mathbb{Z}_2, *)) \simeq \{0, \omega\}, \quad (1.6)$$

as a discrete 2-type. Here ω is the Schur class of

$$1 \longrightarrow \mathbb{Z}_2 \longrightarrow 2A_4 \cong SL(2, 3) \longrightarrow A_4 \longrightarrow 1. \quad (1.7)$$

Every pointed characteristic law induced by A_4 is equivalent to 0 or $\omega(A)$ in the original source coordinate. The calculation reduces to

$$H^1(BA_4; \mathbb{Z}_2) = 0, \quad H^2(BA_4; \mathbb{Z}_2) = \mathbb{Z}_2 \langle \omega \rangle. \quad (1.8)$$

The mixed group $H^1(BA_4) \otimes H^1(BO(2))$ vanishes with the first factor.

The tangent classes remain visible in the calculation. On $\mathbb{RP}^2$, let the generator holonomy of the A_4 bundle be a nontrivial element of its normal Klein subgroup. Then

$$w_1(TM)^2 = w_2(TM) = \omega(A) \neq 0. \quad (1.9)$$

On an F -face triangulation at positive gauge coupling, the two K -flux sectors differ by $(2 \sinh J_g)^F$. The partition function detects the ordinary component on this surface, where the tangent classes are nonzero.

2 The microscopic class

The classification uses one microscopic model and a specified family of source-covariant presentations.

2.1 Tangential structure, gauge field, and matter

Spacetime is a bosonic 1 + 1-manifold whose only tangential datum is an $O(2)$ structure. Let e be a transition cocycle for a rank-two $O(2)$ bundle E . On a branched triangulation, its determinant gives

$$w \in Z^1(M; \mathbb{Z}_2), \quad [w] = w_1(E), \quad (2.1)$$

and a finite-star representative gives $w_2(E) \in Z^2(M; \mathbb{Z}_2)$. If η is a zero-cochain of local frame changes, denote the transformed transition cocycle by e^η . Its determinant changes by $w \mapsto w + \delta \bar{\eta}$, with $\bar{\eta} \in C^0(M; \mathbb{Z}_2)$. We impose the formal source identities for arbitrary E and specialize to $E = TM$ afterward.

The gauge Hilbert space contains one qubit (X_ℓ^g, Z_ℓ^g) on each spatial link. The dynamical gauge group is $\mathbb{Z}_2$, with hard Gauss law

$$Q_v = X_{v-1/2}^g X_{v+1/2}^g = 1. \quad (2.2)$$

All dynamical matter is gauge neutral.

Each vertex carries a neutral four-state spin

$$\mathcal{H}_v^m = \text{span}_{\mathbb{C}}\{|1\rangle, |2\rangle, |3\rangle, |4\rangle\}. \quad (2.3)$$

The ordinary symmetry $G = A_4$ acts by even permutations,

$$U_g = \bigotimes_v P_g, \quad P_g|r\rangle = |g(r)\rangle. \quad (2.4)$$

Local gauge-invariant projectors detect this action faithfully. The only proper nontrivial normal subgroup of A_4 is V_4 , and $A_4/V_4 \cong \mathbb{Z}_3$. Every nonidentity element of V_4 moves one of the projectors, so the physical action remains faithful to A_4 . Its matter support also separates it from the link gauge transformation.

Spatial reflection permutes sites and links and fixes the four internal labels. Complex conjugation in this basis supplies the real structure. We calibrate the source as a direct product: reflection fixes the A_4 label, A_4 commutes with K , and the bare mixed junction carries the identity K -point. Composing reflection with the outer automorphism of A_4 gives another source problem.

2.2 Exact one-form symmetry and the controlled Hamiltonian

The point operator in (1.3) is movable because (2.2) identifies neighboring X_ℓ^g 's on the physical Hilbert space. Its charged line is

$$W(\gamma) = (-1)^{\int_\gamma a}, \quad W(S^1) = \prod_\ell Z_\ell^g. \quad (2.5)$$

With $I(x, \gamma)$ the mod-two intersection number, the fusion and action are

$$\mathcal{U}_K^2 = W^2 = 1, \quad \mathcal{U}_K(x)W(\gamma) = (-1)^{I(x, \gamma)}W(\gamma)\mathcal{U}_K(x). \quad (2.6)$$

The local operator algebra proves that K is exact. Lift a finite-support density to the universal-cover chain, expand it in Pauli monomials, and mark every link carrying Z^g or Y^g . Gauge invariance gives even marked incidence at each vertex. The chain supports only the empty finite mod-two one-cycle. Each gauge-link factor is I or X^g , so every local gauge-invariant operator commutes with $\mathcal{U}_K$. Neutral matter carries zero Wilson endpoint charge.

Define

$$|+\rangle = \frac{1}{2} \sum_{r=1}^4 |r\rangle, \quad P_+ = |+\rangle\langle+|. \quad (2.7)$$

The controlled Hamiltonian is

$$H_* = -h_g \sum_\ell X_\ell^g + h_m \sum_v (1 - P_{+,v}), \quad h_g, h_m > 0. \quad (2.8)$$

The matter ground state is the unique product of $|+\rangle$'s and has gap h_m . On an L -link circle, Gauss law leaves the two uniform sectors $X_\ell^g = q = \pm 1$. The $q = -1$ sector costs $2h_g L$, so the combined gap is bounded below by $\min(h_m, 2h_g)$, independently of L .

2.3 Definition of the certified class

Definition 2.1. An object

$$(H_T, V, \mathfrak{Z}_T) \in \widehat{\mathcal{C}}_{O, A_4}^{\text{cert, abs}}$$

contains the following data.

1. V is a finite-depth circuit of finite-range gates on the universal-cover chain, with depth and range independent of system size and subdivision. Its gates are real and reflection equivariant. They commute with Q_v and U_g , and fix every X_ℓ^g and the winding Wilson line W .

2. The Hamiltonian is exactly

$$H_T = V H_* V^{-1}. \quad (2.9)$$

Equation (2.9) is the full Hamiltonian list for the class.

3. A base completion $\mathfrak{Z}_*$ is an absolute bounded-star symmetric monoidal tensor functor natural in independent (A, E, B) . Its bulk tensors, boundary maps, defect junctions, source morphisms, and gauge-for-gauge morphisms share a uniform finite-star radius. Before the specialization $E = TM$, they obey restriction, source-gauge equivariance, collar contraction, and morphism composition as identities for every formal source. Rebranching and common refinement commute with the source labels, defect OPE, and graph-comparison maps. A regulator comparison carries transformed couplings, local factorizations, and positive normalizers.
4. Conjugating every tensor of $\mathfrak{Z}_*$ by the same V gives $\mathfrak{Z}_T$. The object includes this source-natural intertwiner for every formal (A, E, B) .

Compatible absolute scalar responses may include A_4 SPT, Euler, Stiefel–Whitney, $(-1)^{\int B}$, and higher-cup representatives. Uncancelled source variation or a different fractionalization defines another class. Section 4 admits a non-scalar graph when it appears as a coherent embedding in the calibrated defect OPE.

The state sum in section 3, with $V = 1$, supplies one object of the class. Conjugation preserves the spectrum of H_* , and the certificate carries the source and defect equivalence.

2.4 Symmetry inventory

The source-free matter projector has an independent per-site stabilizer

$$\prod_v (U(1) \times U(3)), \quad (2.10)$$

preserving $\mathbb{C}|+\rangle \oplus |+\rangle^\perp$. Modulo scalar actions, each faithful factor is $U(3)$, which contains S_4 . A certified circuit transports this algebra to a finite-depth dressed one. The same source-free Hamiltonian admits a diagonal S_4 source and a reflection composed with an odd permutation. These are different calibrated source problems from the direct-product A_4 source.

The movable gauge-point algebra exceeds the selected two-object subgroup. Gauss law reduces any product of X^g 's to 1 or a single X_ℓ^g , and mobility through arbitrary matter states makes the matter factor scalar. The point algebra is

$$\mathcal{A}_{\text{top}} = \text{span}_{\mathbb{C}}\{1, X_\ell^g\} \cong \mathbb{C} \oplus \mathbb{C}. \quad (2.11)$$

Its primitive idempotents and vacuum-normalized unitaries are

$$\Pi_{\pm} = \frac{1 \pm X_\ell^g}{2}, \quad \mathcal{U}_\theta = \Pi_+ + e^{i\theta}\Pi_-. \quad (2.12)$$

The full unitary group of $\mathcal{A}_{\text{top}}$ is $U(1) \times U(1)$. Its scalar phase acts trivially by conjugation. Fixing its value on the $q = +1$ vacuum leaves an accidental $U(1)^{(1)}$. The selected subgroup is

$$\{\mathcal{U}_0, \mathcal{U}_\pi\} = \{1, X_\ell^g\} \cong \mathbb{Z}_2^{(1)}. \quad (2.13)$$

Writing $W_{+-} = \Pi_+ W \Pi_-$ and $W_{-+} = \Pi_- W \Pi_+$, we find

$$\mathcal{U}_\theta W_{+-} \mathcal{U}_\theta^{-1} = e^{-i\theta} W_{+-}, \quad \mathcal{U}_\theta W_{-+} \mathcal{U}_\theta^{-1} = e^{i\theta} W_{-+}. \quad (2.14)$$

Every certified V fixes and transports this algebra. The unoriented real structure sends $\mathcal{U}_\theta \mapsto \mathcal{U}_{-\theta}$, which gives the larger $U(1)$ an inversion local system. Our classification fixes the pointwise subgroup (2.13).

3 Regulator and gluing

At fixed cellulation, the state sum realizes source transformations and boundary contractions as identities.

3.1 State sum and gauge transformations

On a branched surface, introduce the finite cochain variables [5],

$$a \in C^1(M; \mathbb{Z}_2), \quad B \in Z^2(M; \mathbb{Z}_2), \quad s_v \in \{1, 2, 3, 4\}. \quad (3.1)$$

A flat A_4 background has edge transports $A_{ij} \in A_4$ satisfying

$$A_{ji} = A_{ij}^{-1}, \quad A_{ij} A_{jk} = A_{ik}, \quad (3.2)$$

on every ordered face. The base weight is

$$f = \delta a + B, \quad S_0 = -J_g \sum_p (-1)^{f_p} - J_m \sum_{(ij)} \delta_{s_i, A_{ij} s_j}. \quad (3.3)$$

It is real and independent of a face orientation. Regulator comparisons allow face couplings $J_{g,p} > 0$ and edge couplings $J_{m,ij} \geq 0$. The displayed action is the uniform specialization.

The source and dynamical transformations are

$$a \longmapsto a + \delta \xi, \quad \text{dynamical gauge,}$$

$$\begin{aligned}
A_{ij} &\mapsto g_i A_{ij} g_j^{-1}, & s_i &\mapsto g_i s_i, & A_4 \text{ source gauge,} \\
e &\mapsto e^\eta, & & & O(2) \text{ frame change,} \\
B &\mapsto B + \delta\Lambda, & a &\mapsto a + \Lambda, & K \text{ source gauge.} \quad (3.4)
\end{aligned}$$

Here $\xi \in C^0(M; \mathbb{Z}_2)$, $\Lambda \in C^1(M; \mathbb{Z}_2)$, $g_i \in A_4$, and η is the zero-cochain of local frame changes defined above. The replacement $\Lambda \mapsto \Lambda + \delta\chi$ is a gauge-for-gauge two-morphism because its action on a is a dynamical gauge transformation.

With $S = S_0$, the closed partition function is

$$Z[M; A, e, B] = \frac{1}{|C^0(M; \mathbb{Z}_2)|} \sum_{a \in C^1(M; \mathbb{Z}_2)} \sum_{s: M^0 \rightarrow \{1,2,3,4\}} e^{-S[a,s;A,e,B]}. \quad (3.5)$$

The denominator is the order of the vertex gauge group, including its constant stabilizer. The sum runs over matter colors. The A_4 field remains a fixed background.

3.2 Hamiltonian transfer matrix

With zero spatial matter coupling, an anisotropic limit of (3.3) gives (2.8). Commutativity makes the transfer relation exact. At trivial temporal holonomy, the four-state Potts kernel is

$$(T_m(\kappa_m))_{ji} = e^{\kappa_m \delta_{ji}} = (e^{\kappa_m} - 1)I_{ji} + (J_4)_{ji} = (e^{\kappa_m} - 1)e_{ji}^{uP_+}, \quad e^u = \frac{e^{\kappa_m} + 3}{e^{\kappa_m} - 1}. \quad (3.6)$$

Here $J_4 = 4P_+$ is the all-ones matrix. Every prescribed $u > 0$ has the positive solution

$$e^{\kappa_m} = \frac{e^u + 3}{e^u - 1}. \quad (3.7)$$

Taking $u = \Delta\tau h_m$ gives the matter term in (2.8) up to a positive scalar. A temporal holonomy replaces this kernel by $T_m(\kappa_m)P_A$, where P_A is its four-state permutation matrix.

For a temporal gauge plaquette,

$$T_g(\kappa_g) = \begin{pmatrix} e^{\kappa_g} & e^{-\kappa_g} \\ e^{-\kappa_g} & e^{\kappa_g} \end{pmatrix} = \sqrt{2 \sinh(2\kappa_g)} e^{u_g X^g}, \quad \tanh u_g = e^{-2\kappa_g}. \quad (3.8)$$

Every prescribed $u_g > 0$ has the positive solution

$$\kappa_g = -\frac{1}{2} \log(\tanh u_g). \quad (3.9)$$

Taking $u_g = \Delta\tau h_g$ gives the electric term. Summing the temporal vertex bits inserts $\prod_v (1 + Q_v)/2$. Restricted to the Gauss-law image, one time slice is a positive scalar times $e^{-\Delta\tau H^*}$. All terms in (2.8) commute, so this equality is exact.

3.3 Refinement and boundary gluing

Finite-coupling weights depend on the regulator. Common refinement uses local tensor factorizations with transformed couplings. For

$$w_J(f) = e^{J(-1)^f}, \quad (3.10)$$

a face split obeys

$$\sum_{x \in \mathbb{Z}_2} w_{J_1}(x) w_{J_2}(x + f) = C(J_1, J_2) w_J(f), \quad \tanh J = \tanh J_1 \tanh J_2, \quad (3.11)$$

with

$$C(J_1, J_2) = 2\sqrt{\cosh(J_1 + J_2) \cosh(J_1 - J_2)}. \quad (3.12)$$

Every positive J has a positive equal split defined by $J_1 = J_2$ and $\tanh J_1 = \tanh J_2 = \sqrt{\tanh J}$.

A matter-edge split is the matrix product

$$T_m(\kappa_1) T_m(\kappa_2) = C_m T_m(\kappa), \quad \frac{e^\kappa + 3}{e^\kappa - 1} = \frac{e^{\kappa_1} + 3}{e^{\kappa_1} - 1} \frac{e^{\kappa_2} + 3}{e^{\kappa_2} - 1}, \quad (3.13)$$

where

$$C_m = \frac{(e^{\kappa_1} - 1)(e^{\kappa_2} - 1)}{e^\kappa - 1}. \quad (3.14)$$

The ratio

$$R(\kappa) = \frac{e^\kappa + 3}{e^\kappa - 1}, \quad (3.15)$$

is a bijection from $(0, \infty)$ to $(1, \infty)$. Every positive κ has a positive equal split determined by $R(\kappa_1) = R(\kappa_2) = \sqrt{R(\kappa)}$.

A stellar face subdivision introduces a center vertex with matter label $r \in \{1, 2, 3, 4\}$. Its matter tensor is the invariant identity insertion

$$I_{\text{stell}}(r) = \frac{1}{4}, \quad \sum_{r=1}^4 I_{\text{stell}}(r) = 1. \quad (3.16)$$

The new spokes carry $J_m = 0$, so summing the center label preserves the boundary-edge Potts tensor. For subdivided fluxes $f_1 + f_2 + f_3 = f$, the spoke-bit quotient is

$$\begin{aligned} S_{\text{stell}}(f) &:= \frac{1}{2} \sum_{a_{0c}, a_{1c}, a_{2c} \in \mathbb{Z}_2} \prod_{q=1}^3 w_{J_q}(f_q) \\ &= \sum_{x, y \in \mathbb{Z}_2} w_{J_1}(x) w_{J_2}(y) w_{J_3}(f + x + y) \\ &= C_{123} w_J(f), \quad \tanh J = \prod_{q=1}^3 \tanh J_q. \end{aligned} \quad (3.17)$$

Here C_{123} is the iterated normalizer from (3.11); the factor $1/2$ contributes the new vertex to the denominator in (3.5). Edge subdivisions use (3.13). The algebraic identities in (3.11), (3.13), and (3.16) make the common-refinement maps associative. Pullback of a background cocycle commutes with every calibrated source tensor.

On a cobordism M with boundary N , divide only by gauge transformations that are trivial on N and retain the boundary fields:

$$Z_M(a_N, s_N) = \frac{1}{|C^0(M, N; \mathbb{Z}_2)|} \sum_{\substack{a|_N = a_N \\ s|_N = s_N}} e^{-S_M}. \quad (3.18)$$

Half of each seam-edge matter weight belongs to each adjacent kernel. If $M = M_1 \cup_N M_2$, then

$$Z_M = \frac{1}{|C^0(N; \mathbb{Z}_2)|} \sum_{a_N, s_N} Z_{M_1}(a_N, s_N) Z_{M_2}(a_N, s_N). \quad (3.19)$$

The normalization follows from

$$|C^0(M)| = |C^0(M_1, N)| |C^0(N)| |C^0(M_2, N)|. \quad (3.20)$$

For a general completion, (3.19) contracts every open collar index and response-state line with the same boundary groupoid measure. Source equivariance on a boundary retains the transformed boundary fields:

$$\begin{aligned} Z_M(g \cdot A, e, B; a_N, g|_N s_N) &= Z_M(A, e, B; a_N, s_N), \\ Z_M(A, e, B + \delta\Lambda; a_N + \Lambda|_N, s_N) &= Z_M(A, e, B; a_N, s_N). \end{aligned} \quad (3.21)$$

The transformations (3.4) give the exact Ward identities

$$Z[M; g \cdot A, e, B] = Z[M; A, e^\eta, B] = Z[M; A, e, B + \delta\Lambda] = Z[M; A, e, B]. \quad (3.22)$$

For the base weight, these are changes of summation variables or literal independence of e . General objects obey them by the absolute membership condition. The same K -background transformation acts on a Wilson line by

$$W(\gamma) \mapsto (-1)^{\int_\gamma \Lambda} W(\gamma). \quad (3.23)$$

An external or boundary endpoint state carries the inverse factor; no dynamical bulk endpoint is introduced.

3.4 Generating source and defect tensors

The base functor has local generators. An A_4 wall labelled by g acts on a matter index by the real permutation matrix P_g . Bare junctions satisfy

$$P_g P_h = P_{gh}. \quad (3.24)$$

The calibrated K point inserts X_ℓ^g . A linking Wilson line inserts Z_ℓ^g and gives the sign in (2.6). Orientation reversal permutes the branched simplex indices and complex-conjugates their real entries. Source changes use (3.4). Collar contraction uses (3.19). A certified circuit V conjugates each open leg and junction to produce $\mathfrak{Z}_T$.

4 Laws from the defect OPE

Defect-junction quotients determine the local background graph.

Let $\text{Disk}_2^{O, \text{reg}}$ be the site of finite unions of stratified two-disks with collars, frame charts, and ordered simplicial regulators. Its morphisms are generated by restriction to stars, collar-preserving embeddings, source isomorphisms, rebranchings, and stellar subdivisions. Define

$$\mathcal{X} = \mathbf{B}A_4 \times \mathbf{B}O(2), \quad \mathcal{K} = \mathbf{B}^2\mathbb{Z}_2. \quad (4.1)$$

The value $\mathcal{X}(U)$ records a flat A_4 bundle and an independent rank-two bundle E . The Picard 2-stack $\mathcal{K}(U)$ contains two-cocycles, their one-cochain gauge morphisms, and zero-cochain gauge-for-gauge morphisms. Physical evaluation follows the formal source identities and then sets $E = TU$.

For a formal-source disk germ (U, A, E) , let $J_{A,E}^{\text{tr}}$ and $J_{A,E}^{\text{bare}}$ be the transported and bare calibrated junction tensors. Their operational quotient is the OPE comparison

$$J_{A,E}^{\text{tr}} \simeq R_{A,E} \otimes J_{A,E}^{\text{bare}}. \quad (4.2)$$

A canonical- K graph embeds $\{1, \mathcal{U}_K\}$ coherently into the OPE represented by $R_{A,E}$, with a linking Wilson line fixing its exponent. The tensors of $\mathfrak{Z}_T$ move and fuse its graph lines. They also supply the source-gauge maps and associator. Spectator multiplicities stay in the ambient OPE, and each coherent embedding defines its own graph. Physical evaluation restricts the graph to $E = TU$.

Lemma 4.1 (OPE reconstruction). *Every coherent operational canonical- K embedding determines a local 2-natural map*

$$\Phi : \mathcal{X} \longrightarrow \mathcal{K}, \quad B = \Phi(A, E), \quad (4.3)$$

unique for that embedding up to a B -gauge one-morphism and a gauge-for-gauge two-morphism.

Proof. Represent the quotient point by $\mathcal{U}_K^\phi$, with $\phi \in \mathbb{Z}_2$. Its linking eigenvalue $(-1)^\phi$ fixes ϕ , even in the presence of spectator multiplicity. The two fusions of three junctions agree by the OPE associator, so $\delta\phi = 0$. A change of source trivialization gives a one-cochain q satisfying $\phi' - \phi = \delta q$. Comparing two successive changes gives a zero-cochain t , and the OPE pentagon gives its gauge-for-gauge coherence.

These identities hold for every formal (A, E, B) germ. Restriction, common refinement, and collar contraction give effective descent, so the local data stackify to (4.3). The OPE supplies the graph, and the Wilson line fixes its point label. $\square$

Let

$$i : \mathbf{BO}(2) \longrightarrow \mathcal{X}, \quad p : \mathcal{X} \longrightarrow \mathbf{BO}(2), \quad (4.4)$$

be the trivial- A_4 inclusion and the projection. We isolate the part induced by A through

$$\Psi = \Phi + p^* i^* \Phi. \quad (4.5)$$

Since the coefficients are mod two, $i^* \Psi \simeq 0$ with its inherited null-homotopy. Equation (4.5) leaves the independent B -coordinate and every pure-source embedding fixed.

5 Classification

The relative mapping space computes the components and lower homotopy groups of (4.5).

5.1 Stackification and the relative mapping space

Set

$$X = BA_4 \times BO(2), \quad Y = \{*\} \times BO(2). \quad (5.1)$$

The selected K has constant coefficients because $\text{Aut}(\mathbb{Z}_2) = 1$. The inversion local system of the accidental $U(1)$ belongs to a separate target.

Definition 5.1. Let $\text{Law}_{A_4, O, \text{cert}, \text{pt}}^2$ contain pointed 2-natural maps $\Psi : \mathcal{X} \rightarrow \mathcal{K}$, with a chosen null-homotopy on $\mathbf{BO}(2)$, together with their gauge and gauge-for-gauge morphisms. We impose *characteristic descent*: bundle pullbacks act naturally, homotopies of classifying maps preserve the isomorphism class, and a universal simplicial representative pulls back to a bounded-star cochain. Its lower morphisms obey the regulator descent conditions of definition 2.1.

Lemma 4.1 and (4.5) place every OPE graph with characteristic descent in this 2-groupoid. One theory may realize only some of its objects.

Lemma 5.2 (Characteristic-law classification). *The ambient 2-groupoid of pointed laws is*

$$\text{Law}_{A_4, O, \text{cert}, \text{pt}}^2 \simeq \tau_{\leq 2} \text{Map}((X, Y), (B^2\mathbb{Z}_2, *)). \quad (5.2)$$

Proof. Characteristic descent makes a law a natural homotopy-invariant assignment to an A_4 bundle and a rank-two $O(2)$ bundle. Their classifying maps land in $X = BA_4 \times BO(2)$. The chosen trivialization at trivial A_4 source is the homotopy fiber of restriction to $Y = \{*\} \times BO(2)$, which gives (5.2).

For an explicit cochain model, use $BA_4 = N_\bullet A_4$, a singular simplicial model for $BO(2)$, and $B^2\mathbb{Z}_2 = \Gamma(\mathbb{Z}_2[2])$. Dold–Kan identifies objects, one-morphisms, and two-morphisms with relative cochains of degrees 2, 1, and 0 [6]. A universal degree-two cocycle evaluates simplexwise. Its pullback and every universal cochain homotopy have bounded-star representatives compatible with the refinement maps. $\square$

Characteristic descent admits tangent-dependent laws. Replacing A_4 by $\mathbb{Z}_2$ produces $a(A)w_1(E)$. For A_4 , the identity $H^1(BA_4; \mathbb{Z}_2) = 0$ removes the mixed class.

5.2 A_4 cohomology and the relative groups

The abelianization and Schur multiplier are

$$H_1(A_4; \mathbb{Z}) \cong \mathbb{Z}_3, \quad H_2(A_4; \mathbb{Z}) \cong \mathbb{Z}_2. \quad (5.3)$$

The universal-coefficient sequence [7] gives

$$0 \longrightarrow \text{Ext}^1(\mathbb{Z}_3, \mathbb{Z}_2) \longrightarrow H^2(BA_4; \mathbb{Z}_2) \longrightarrow \text{Hom}(\mathbb{Z}_2, \mathbb{Z}_2) \longrightarrow 0. \quad (5.4)$$

The Ext term vanishes and the last term is $\mathbb{Z}_2$. Since $\text{Hom}(A_4, \mathbb{Z}_2) = 0$, the sequence gives

$$H^1(BA_4; \mathbb{Z}_2) = 0, \quad H^2(BA_4; \mathbb{Z}_2) = \mathbb{Z}_2 \langle \omega \rangle. \quad (5.5)$$

The Lyndon–Hochschild–Serre sequence gives a second calculation. Write $A_4 = V_4 \rtimes \mathbb{Z}_3$. Since 3 is invertible in $\mathbb{F}_2$, the sequence reduces to invariants,

$$H^*(BA_4; \mathbb{F}_2) \cong H^*(BV_4; \mathbb{F}_2)^{\mathbb{Z}_3}. \quad (5.6)$$

If $H^*(BV_4; \mathbb{F}_2) = \mathbb{F}_2[u, v]$ and a generator sends $u \mapsto v$, $v \mapsto u + v$, then the degree-one invariant subspace vanishes and degree two has the single invariant

$$u^2 + uv + v^2. \quad (5.7)$$

Its restriction to each involution subgroup is t^2 , so it is the nonzero binary tetrahedral class.

Since

$$H^*(BO(2); \mathbb{Z}_2) = \mathbb{Z}_2[w_1, w_2], \quad (5.8)$$

the Künneth theorem gives

$$H^2(X; \mathbb{Z}_2) = \text{span}_{\mathbb{Z}_2}\{\omega, w_1^2, w_2\}. \quad (5.9)$$

The mixed summand is absent because $H^1(BA_4; \mathbb{Z}_2) = 0$.

Restriction $H^1(X) \rightarrow H^1(Y)$ is an isomorphism. In degree two, its kernel is generated by ω , and its image contains w_1^2 and w_2 . The relative long exact sequence gives

$$H^2(X, Y; \mathbb{Z}_2) = \mathbb{Z}_2\langle\omega\rangle, \quad H^1(X, Y; \mathbb{Z}_2) = H^0(X, Y; \mathbb{Z}_2) = 0. \quad (5.10)$$

For an Eilenberg–Mac Lane target, $\pi_j \text{Map}((X, Y), (K(\mathbb{Z}_2, 2), *)) = H^{2-j}(X, Y; \mathbb{Z}_2)$. The law 2-type is the discrete pair $\{0, \omega\}$.

5.3 Fixed-calibration classification theorem

Theorem 5.3 (Fixed-calibration law classification). *Let $(H_T, V, \mathfrak{Z}_T) \in \widehat{\mathcal{C}}_{O, A_4}^{\text{cert, abs}}$ carry a local defect OPE graph Φ into the calibrated canonical $K = \mathbb{Z}_2^{(1)}$, whose pointed part has characteristic descent as defined in definition 5.1. Then its A_4 -induced pointed law is equivalent, in the original fixed source 2-groupoid, to*

$$B = c\omega(A), \quad c \in \mathbb{Z}_2. \quad (5.11)$$

Its pointed characteristic class has vanishing mixed tangential component.

Proof. Apply lemma 4.1 to the coherent canonical- K quotient embedding, then use lemma 5.2 to place its pointed difference in the relative mapping 2-type. Equation (5.10) identifies this type with the ordinary pointed mapping 2-type from BA_4 . A universal finite-star relative one-cochain L satisfies

$$\Psi(A, E) = c\omega(A) + \delta L(A, E). \quad (5.12)$$

Together with its two-morphism descent, the B -gauge homotopy L preserves the microscopic A_4 action, $\mathcal{U}_K$, W , and every calibrated junction. The supplied source-natural V transports this comparison through every completed presentation in the class. $\square$

This classification precedes the specialization $E = TM$ and any surface Wu reduction. Pure tangent points may remain in the unpointed origin $i^*\Phi$; taking the pointed difference removes them.

6 The cocycle and its defects

The binary tetrahedral extension supplies a cochain representative of the ordinary component and its source-gauge descent.

6.1 Binary tetrahedral cocycle

Choose a normalized section $\sigma : A_4 \rightarrow \tilde{A}_4 = 2A_4$, with $\sigma(1) = 1$, and define $\omega(g, h) \in \mathbb{Z}_2$ by

$$\sigma(g)\sigma(h) = z^{\omega(g,h)}\sigma(gh), \quad z^2 = 1. \quad (6.1)$$

Associativity gives

$$\omega(h, k) + \omega(g, hk) = \omega(g, h) + \omega(gh, k). \quad (6.2)$$

On an ordered flat triangle,

$$\omega(A)_{012} = \omega(A_{01}, A_{12}). \quad (6.3)$$

For each positive edge $i < j$, choose a lift $\tilde{A}_{ij}$ and set $\tilde{A}_{ji} = \tilde{A}_{ij}^{-1}$. The face curvature is

$$\tilde{A}_{01}\tilde{A}_{12}\tilde{A}_{02}^{-1} = z^{\omega(A)_{012}}. \quad (6.4)$$

Changing edge lifts by $\tilde{A}_{ij} \mapsto z^{\Lambda_{ij}}\tilde{A}_{ij}$ sends $\omega(A) \mapsto \omega(A) + \delta\Lambda$.

For an A_4 gauge transformation $A_{ij} \mapsto g_i A_{ij} g_j^{-1}$, the canonical vertex lifts give

$$\sigma(g_i)\sigma(A_{ij})\sigma(g_j)^{-1} = z^{q_g(A)_{ij}}\sigma(g_i A_{ij} g_j^{-1}). \quad (6.5)$$

Taking the central face curvature yields

$$\omega(g \cdot A) = \omega(A) + \delta q_g(A). \quad (6.6)$$

The replacement $\tilde{g}_i = z^{r_i}\sigma(g_i)$ sends $q_g \mapsto q_g + \delta r$, which is the corresponding gauge-for-gauge change.

If g is followed by h , define

$$\sigma(h_i)\sigma(g_i) = z^{t_{h,g,i}}\sigma(h_i g_i). \quad (6.7)$$

Then

$$q_g(A) + q_h(g \cdot A) = q_{hg}(A) + \delta t_{h,g}, \quad (6.8)$$

$$t_{k,h} + t_{kh,g} = t_{h,g} + t_{k,hg}. \quad (6.9)$$

A different section $\sigma'(g) = z^{\ell(g)}\sigma(g)$ sends $\omega \mapsto \omega + \delta\ell$.

6.2 Wall junctions and orientation defects

A flat A_4 background is a network of codimension-one walls [8]. Crossing a g -wall permutes the local projectors by (2.4). Bare walls fuse by $g, h \mapsto gh$. The ordinary law ω decorates an ordered g, h junction by

$$\mathcal{U}_K^{\omega(g,h)}. \quad (6.10)$$

Equation (6.2) is the three-wall associator. The decorated labels multiply as

$$(g, \epsilon)(h, \epsilon') = (gh, \epsilon + \epsilon' + \omega(g, h)). \quad (6.11)$$

Take $g = (12)(34)$ and $h = (13)(24)$. The preimage in $Q_8 \subset 2A_4$ of this commuting V_4 pair has anticommuting lifts. Their commutator gives

$$\omega(g, h) + \omega(h, g) = 1. \quad (6.12)$$

The crossing of these commuting walls carries one net K -point, and a Wilson line detects its sign.

The orientation wall is

$$D_O = \text{PD}(w_1(TM)). \quad (6.13)$$

The point w_1^2 is its mod-two self-intersection. For a general rank-two E , $w_2(E)$ is the zero locus of a generic section. Both points occur already at trivial A_4 source and cancel in the pointed law (4.5). A mixed point at an A_4 -wall and orientation-wall intersection would require a character $A_4 \rightarrow \mathbb{Z}_2$. The identity $\text{Hom}(A_4, \mathbb{Z}_2) = 0$ rules it out.

7 Partition functions and the $\mathbb{RP}^2$ test

Equation (3.22) identifies the unpointed graph with its ordinary representative. We evaluate this identity on surfaces with nonzero tangent classes.

7.1 Exact partition comparison

Set

$$\phi_0 = i^* \Phi, \quad \Phi_0 = p^* \phi_0. \quad (7.1)$$

Equation (5.12) gives

$$\Phi(A, E) = \Phi_0(E) + c\omega(A) + \delta L(A, E). \quad (7.2)$$

The Ward identity (3.22) gives, for every independent external K -source and every closed $O(2)$ surface,

$$Z_T[M; A, e, B_{\text{ext}} + \Phi(A, E)] = Z_T[M; A, e, B_{\text{ext}} + \Phi_0(E) + c\omega(A)]. \quad (7.3)$$

The pure tangent origin Φ_0 remains on both sides. On a cobordism, the boundary restriction of L supplies the K -background intertwiner; its zero-cochain descent controls composition. Equation (7.3) then extends to boundary tensors, Wilson-line correlators, and defect junctions.

The base completion has zero anomaly. Its onsite A_4 action and $\delta a + B$ coupling define the source response directly on $O(2)$ surfaces. Every certified object inherits the corresponding Ward identities. Scalar response phases multiply both sides of (7.3) by the same factor and leave the graph fixed.

7.2 Finite-coupling flux sectors

At the auxiliary flat-projector point, the gauge factor is

$$\begin{aligned} Z_{\text{gauge}}^{\text{fp}}(M; B) &= \frac{1}{|C^0(M; \mathbb{Z}_2)|} \sum_{a \in C^1(M; \mathbb{Z}_2)} \prod_p \delta_2((\delta a + B)_p) \\ &= \begin{cases} |H^1(M; \mathbb{Z}_2)| / |H^0(M; \mathbb{Z}_2)|, & [B] = 0, \\ 0, & [B] \neq 0. \end{cases} \end{aligned} \quad (7.4)$$

Here δ_2 is the indicator of zero in $\mathbb{Z}_2$. We use (7.4) only as an algebra check; the certified Hamiltonian has finite h_g .

For a connected closed triangulation with F faces and uniform $J_g > 0$, the finite-coupling gauge factor is

$$Z_{\text{gauge}}^{J_g}(M; B) = \frac{|H^1(M; \mathbb{Z}_2)|}{|H^0(M; \mathbb{Z}_2)|} \frac{(2 \cosh J_g)^F + (-1)^{\int_M B} (2 \sinh J_g)^F}{2}. \quad (7.5)$$

The image of $\delta : C^1 \rightarrow C^2$ is the even-parity subspace, with a constant number of preimages for each flux in a given parity sector. Face-dependent couplings replace the two powers in (7.5) by $\prod_p 2 \cosh J_p$ and $\prod_p 2 \sinh J_p$.

7.3 Simultaneously nonzero tangent and ordinary classes

Let x generate $H^1(\mathbb{RP}^2; \mathbb{Z}_2)$. The stable identity [9]

$$T\mathbb{RP}^2 \oplus \mathbf{1} \cong 3\gamma, \quad (7.6)$$

for the tautological line γ gives

$$w(T\mathbb{RP}^2) = (1 + x)^3 = 1 + x + x^2. \quad (7.7)$$

The tangent classes are

$$w_1(TM) = x, \quad w_1(TM)^2 = w_2(TM) = x^2, \quad \int_{\mathbb{RP}^2} x^2 = 1. \quad (7.8)$$

Map the generator of $\pi_1(\mathbb{RP}^2) = \langle u \mid u^2 = 1 \rangle$ to an involution $r \in V_4 \subset A_4$. The inverse image of V_4 in $2A_4$ is Q_8 , so every lift satisfies

$$\tilde{r}^2 = z. \quad (7.9)$$

The pulled-back extension is $\mathbb{Z}_2 \rightarrow \mathbb{Z}_4 \rightarrow \mathbb{Z}_2$, the nonzero class in $H^2(B\mathbb{Z}_2; \mathbb{Z}_2)$, so

$$\int_{\mathbb{RP}^2} \omega(A_r) = 1. \quad (7.10)$$

Since $|H^1(\mathbb{RP}^2; \mathbb{Z}_2)|/|H^0(\mathbb{RP}^2; \mathbb{Z}_2)| = 1$, equation (7.5) gives

$$Z_{\text{gauge}}^{J_g}(\mathbb{RP}^2; 0) - Z_{\text{gauge}}^{J_g}(\mathbb{RP}^2; \omega(A_r)) = (2 \sinh J_g)^F \neq 0. \quad (7.11)$$

The matter factor at fixed A_r is positive and independent of B . The full partition function distinguishes the two sectors on the same surface for which w_1^2 and w_2 are nonzero.

7.4 A nonzero ordinary fractionalization completion

Define

$$Z_{(T_*, \omega)}[M; A, e, B_{\text{ext}}] := Z_{T_*}[M; A, e, B_{\text{ext}} + \omega(A)]. \quad (7.12)$$

Keep the bare wall junction fixed and define the transported junction by

$$J_{g,h}^\omega = \mathcal{U}_K^{\omega(g,h)} \otimes J_{g,h}^{\text{bare}}. \quad (7.13)$$

The lift equations (6.2) and (6.6)–(6.9) supply source and fusion descent. Equations (7.12) and (7.13) define the operational quotient OPE. Pulling back the remaining tensors from the absolute completion gives $\mathfrak{Z}_*^\omega$. The triple $(H_*, 1, \mathfrak{Z}_*^\omega)$ satisfies the membership test and realizes a nonzero ordinary graph. Its microscopic Hamiltonian carries A_4 and K at every scale. The graph records fractionalization data.

8 Scope

Definitions 2.1 and 5.1 set the theorem’s domain: finite-depth certified Hamiltonians whose pointed graphs have characteristic descent. Gauge-charged matter and anomalous source completions require new microscopic data. Frozen relative boundaries change the comparison, as do alternative tangential structures or symmetry targets. The accidental $U(1)^{(1)}$ belongs to an inversion-twisted target. A diagonal S_4 source and an outer-twisted reflection source use different source stacks.

The UV-to-IR benchmark of ref. [1] asks for a microscopic theory with G in the UV and K in the IR. The present Hamiltonian carries A_4 and K at every scale. The theorem classifies its fixed-calibration background laws. The dynamical transmutation benchmark remains open.

Equation (5.12) stays in the original source coordinate. The cochain L is a B -gauge homotopy within one source object, with the $B = 0$ section and the microscopic calibrations fixed. Scalar counterterms act on response phases. The graph classification is off shell.

The stated requirements lead to a short minimality test. Dimension two is the first spacetime dimension with a one-form background; $O(2)$ gives the least restrictive bosonic tangential structure. We take $H = K = \mathbb{Z}_2$, the smallest nontrivial gauge and calibrated higher-form groups. The ordinary group must satisfy

$$H^1(BG; \mathbb{Z}_2) = 0, \quad H^2(BG; \mathbb{Z}_2) \neq 0. \quad (8.1)$$

Odd-order groups have vanishing positive-degree mod-two cohomology. Every even group of order below 12 has a $\mathbb{Z}_2$ quotient. The orders 2, 4, 8 give nontrivial 2-groups; groups of order 6 or 10 have a normal odd Sylow subgroup of index two. The group A_4 , of order 12, is the first finite group satisfying (8.1). Its natural four-state action is the smallest faithful permutation representation. A faithful three-dimensional real representation also exists.

9 Conclusion

Characteristic descent identifies each pointed graph with a map from the pair $(BA_4 \times BO(2), \{*\} \times BO(2))$ to $B^2\mathbb{Z}_2$. Its relative mapping 2-type has components 0 and ω , with trivial lower homotopy groups. The $\mathbb{RP}^2$ calculation separates the ω flux sector at finite coupling on a surface with $w_1^2 = w_2 \neq 0$. The model carries A_4 and K at every scale. The calculation concerns fractionalization data.

A Finite certificates

The program `verify_no_go.js` checks the local Pauli algebra and the calibrated subgroup. The A_4 cocycle and its descent are checked by `verify_a4_no_go.js`. Direct cochain and Boltzmann sums on $\mathbb{RP}^2$ are handled by `verify_rp2_simplicial.js`. The companion text and regulator formulas are checked by `verify_solution_structure.js`.

Script	Checked object	Reported count
<code>verify_no_go.js</code>	Seven-link Pauli algebra, Gauss span, and Wilson charges	16,384 strings; 0 local K violations
<code>verify_a4_no_go.js</code>	$SL(2, 3)$, its A_4 quotient, the normalized bar complex, and source descent	24 lifts; $\dim H^2 = 1$; 3,538,944 cocycle identities
<code>verify_rp2_simplicial.js</code>	Cohomology, cup square, gauge orbits, and finite-coupling sums	6 vertices; 5 + 1 coupling vectors
<code>verify_solution_structure.js</code>	Scope markers, references, refinement, and transfer kernels	42 equation tags; 64 stellar matter checks

Table 1. Finite certificates used with the analytic proof. Exact enumerations use finite arithmetic; sampled phase and Boltzmann identities use the tolerances reported by each script.

The mapping-space classification is analytic.

From the `ai-research` workspace directory, all certificates run with

```
node beyond-cohomology/verify_no_go.js
node beyond-cohomology/verify_a4_no_go.js
node beyond-cohomology/verify_rp2_simplicial.js
node beyond-cohomology/verify_solution_structure.js
```

References

- [1] N. Seiberg and S. Seifnashri, *Symmetry transmutation and anomaly matching*, *JHEP* **09** (2025) 014 [[2505.08618](#)].
- [2] D. Gaiotto, A. Kapustin, N. Seiberg and B. Willett, *Generalized global symmetries*, *JHEP* **02** (2015) 172 [[1412.5148](#)].
- [3] A. Kapustin and R. Thorngren, *Higher symmetry and gapped phases of gauge theories*, in *Algebra, Geometry, and Physics in the 21st Century*, vol. 324 of *Progress in Mathematics*, pp. 177–202, Birkhäuser (2017), [DOI \[1309.4721\]](#).

- [4] A. Kapustin, *Symmetry protected topological phases, anomalies, and cobordisms: Beyond group cohomology*, [1403.1467](#).
- [5] R. Dijkgraaf and E. Witten, *Topological gauge theories and group cohomology*, *Commun. Math. Phys.* **129** (1990) 393.
- [6] P.G. Goerss and J.F. Jardine, *Simplicial Homotopy Theory*, Birkhäuser, Basel (1999), [10.1007/978-3-0348-8707-6](#).
- [7] K.S. Brown, *Cohomology of Groups*, vol. 87 of *Graduate Texts in Mathematics*, Springer, New York (1982), [10.1007/978-1-4684-9327-6](#).
- [8] M. Barkeshli, P. Bonderson, M. Cheng and Z. Wang, *Symmetry fractionalization, defects, and gauging of topological phases*, *Phys. Rev. B* **100** (2019) 115147 [[1410.4540](#)].
- [9] A. Hatcher, *Algebraic Topology*, Cambridge University Press, Cambridge (2002).